

LOADZONE · REQUIREMENTS

LoadZone requirements

Open Hardware Portfolio · amishchadha.com

DOCUMENT CONTROL

Document ID	LDZ-REQ-001	Version	0.2
Status	Draft	Date	2026-09-25
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 2 Technology concept formulated		
Repository	github.com/BoujeeEnjinia1701/loadzone		
Source	docs/03-requirements.md @ 4cdd8a2		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept
0.1	2026-09-25	Amish Chadha	Initial scaffold

LoadZone requirements

These are first-pass requirements for the concept. Targets are proposals for review, awaiting Amish, and will be checked by calculation at TRL 3 and by test at TRL 4 or later. "Status" gives the position of the TRL 2 concept on first-order estimates.

Table 1. Requirements

ID	Requirement	Target	Status at TRL 2	Verification
R1	Detect the state of each vehicle slot	97 % or more of slot states correct over a day, for cars, vans and box trucks, with traffic passing in the next lane	Unverified; magnetometer-only detection may be fooled by adjacent-lane traffic and by high-clearance trucks	Field trial against a manual count or time-lapse survey (TRL 5)
R2	Report changes quickly	Slot state at the server within 60 s of a vehicle arriving or leaving	Met by estimate: 15 s debounce plus under 5 s to send	Timing calculation (TRL 3)
R3	Long battery life	5 years or more at 100 state changes a day plus hourly heartbeats	Met by estimate: about 7 years on two AA-size Li-SOCl ₂ cells	Power budget calculation (TRL 3)
R4	Reach a gateway from road level	Uplinks received by a gateway 1 km away in a street canyon, with a van parked over the puck	Unverified; a vehicle overhead may add 10 to 20 dB of loss (estimate)	Link budget (TRL 3), then field test
R5	Stay within network fair use	30 s or less of uplink airtime per puck per day (TTN fair use)	Partly met: about 9 s a day at SF7 and about 31 s at SF9, so at SF9 or slower the heartbeat must drop to every 2 h	Airtime calculation (TRL 3)
R6	Survive traffic	Withstand a 50 kN static wheel load and repeated drive-overs by loaded trucks without cracking or debonding (target, basis to be set at TRL 3)	Not met for a printed prototype housing; cast polyurethane over potted electronics is proposed but unverified	Load calculation (TRL 3), then load test
R7	Safe, low profile	Height 35 mm or less above the road, rounded edges, high-visibility color	Met in the massing model: 31 mm high, 150 mm diameter, yellow dome	Massing model and design review
R8	Weather and temperature	Sealed to IP68; operate from -25 to +70 °C road surface temperature; resist salt, oil and fuel	Unverified; cell and sensor ratings to be checked against +70 °C	Datasheet review (TRL 3)
R9	Privacy by design	Only slot state, timestamps and device health leave the device; no images, audio or vehicle identifiers	Met by design: a magnetometer cannot read plates or faces	Design review
R10	Open data	Bay state available through an open API that maps onto the Curb Data Specification Events and Metrics APIs (Open Mobility Foundation)	Met in concept; mapping not yet written	Design review (TRL 3)
R11	Sign legible to drivers (option)	Free-slot count legible at 25 m by day and night	Not met: a 7.5 in e-paper panel gives about 80 mm digits, legible at about 10 m by day (estimate), and it is unlit at night	Legibility calculation (TRL 3)
R12	Sign power (option)	Sign runs within the host FieldNode's sensor energy allowance of about 2.75 Wh a day	Met by estimate: about 0.4 Wh a day without night lighting	Power budget (TRL 3)
R13	Low cost	Two-puck bay kit \$120 or less in parts at quantity 1	Met: about \$110. With the sign option, about \$207 plus a host	Priced BOM

ID	Requirement	Target	Status at TRL 2	Verification
			FieldNode (about \$126), which exceeds the budget	
R14	Quick install without road works	One puck bonded in 15 min or less by a two-person crew; removable without damaging the road	Unverified	Install trial (TRL 4 or later)
R15	Tamper and theft resistant	No external screws; removal needs a heat gun or scraper; device reports removal	Unverified; removal detection from a sudden field and orientation change is proposed	Design review (TRL 3)

■ Requirements not met at TRL 2

- **R6 (traffic loads)** is not met by a printed housing and is unverified for cast polyurethane.
- **R11 (sign legibility)** is not met with a 7.5 in e-paper panel; a larger panel, a lit display or relying on a map layer instead are open options.
- **R13 (cost)** is met for the two-puck kit but not with the sign option.
- **R5 (airtime)** is met only at fast data rates unless the heartbeat interval is lengthened.
- **R1 and R4** are the main technical risks and cannot be confirmed without field data.

■ Assumptions

- A vehicle slot is about 7 m long and the puck sits near its center, 1.3 m from the curb face (estimates; bay layouts vary by city).
- Loading bay turnover of up to 50 vehicles a slot a day, or 100 state changes, based on most delivery stops lasting 30 minutes or less ([Urban Freight Lab, 2019](#)).
- Each LoRaWAN uplink costs about 16 mA·s at SF9, as estimated in FieldNode's design precis.
- Wheel load of 50 kN is a placeholder for a loaded truck's heaviest single wheel; the governing load case is to be set at TRL 3.